Arista Access Node HSR-ACC-27 SOP for QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) on ALARM_6652

Node: HSR-ACC-27 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_6652

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) failure on Arista Access Node HSR-ACC-27 at HSR Layout Access. The failure domain occurs due to the inability of the node's QoS policy to effectively manage traffic congestion, leading to microbursts, interface tail drops, and QoS queue starvation. This is a critical issue that can impact network performance and reliability. The underlying protocol mechanics involve the node's QoS policy engine, which is responsible for managing traffic flow based on priority, rate limiting, and policing. However, in this scenario, the policy engine fails to effectively manage traffic congestion, resulting in the observed symptoms. The hardware mechanics involve the node's interfaces, which are experiencing high levels of congestion, leading to microbursts and interface tail drops. The QoS queue starvation is a result of the node's QoS policy engine failing to allocate sufficient buffer space to handle the incoming traffic.

Pre-requisite Checks

- Ensure that the remote hands are dispatched to HSR Layout Access to perform physical checks on the node.

- Verify the status of the redundant link to ensure it is operational and not contributing to the congestion event.

- Ensure that the node's QoS policy is correctly configured and is not causing the congestion event.

Potential Root Causes

Cause	Probability
QoS policy engine failure to manage traffic congestion	High
Interface congestion leading to microbursts and interface tail drops	Medium
QoS queue starvation due to insufficient buffer space	Low

Diagnostic Commands & Expected Outputs

```
Command: show queue queue, Expected Output: Queue Name Queue ID Type
State Drop Tx Rx Cpu Pkt Avg Max Min Mcast Dropped Dropped Dropped
Dropped Dropped Dropped Dropped Dropped Dropped Dropped Dropped
Dropped Dropped Dropped Dropped Dropped, Description: This command
displays the current state of the QoS queues, including the number of
dropped packets, transmitted packets, and received packets. This
```

```
information can help identify if the QoS policy engine is effectively
managing traffic congestion.

Command: show interface Ethernet counters, Expected Output: Interface:
Ethernet , Index 1 (SNMP ifIndex 1), Description: This command
displays the interface counters, including the number of packets
transmitted and received, and the number of packets dropped. This
information can help identify if the interface is experiencing high
levels of congestion.

Command: show qos policy , Expected Output: Policy Name Match Action
Priority Rate Burst Policed Dropped, Description: This command
displays the QoS policy configuration, including the match criteria,
action, priority, and rate limiting. This information can help verify
if the QoS policy is correctly configured and is not causing the
congestion event.
```

Resolution Steps

Step 1: Step: Verify the QoS policy configuration, Command: show qos policy , Description: Verify that the QoS policy is correctly configured and is not causing the congestion event. Check the match criteria, action, priority, and rate limiting to ensure they are set to the desired values.

Step 2: Step: Adjust the QoS policy configuration, Command: config qos policy match action priority rate burst , Description: Adjust the QoS policy configuration to address the congestion event. Adjust the match criteria, action, priority, and rate limiting as necessary to ensure the policy engine can effectively manage traffic congestion.

Step 3: Step: Save the QoS policy configuration, Command: copy running-config startup-config, Description: Save the QoS policy configuration to ensure it persists across reboots.

Post-Check Verification

```
Command: show queue queue, Expected Output: Queue Name Queue ID Type
State Drop Tx Rx Cpu Pkt Avg Max Min Mcast Dropped Dropped Dropped
Dropped Dropped Dropped Dropped Dropped Dropped Dropped Dropped
Dropped Dropped Dropped Dropped Dropped, Description: Verify that the
QoS policy engine is now effectively managing traffic congestion by
displaying the current state of the QoS queues.

Command: show interface Ethernet counters, Expected Output: Interface:
Ethernet , Index 1 (SNMP ifIndex 1), Description: Verify that the
interface is no longer experiencing high levels of congestion by
displaying the interface counters.

Command: show qos policy , Expected Output: Policy Name Match Action
Priority Rate Burst Policed Dropped, Description: Verify that the QoS
policy configuration is correctly updated by displaying the QoS policy
configuration.
```

Roll-back Procedures

- Step: Restore the original QoS policy configuration, Command: copy startup-config running-config, Description: Restore the original QoS policy configuration to ensure the node returns to its previous state.

- Step: Verify the original QoS policy configuration, Command: show qos policy , Description: Verify that the original QoS policy configuration is correctly restored by displaying the QoS policy configuration.